Objects in action

Today

Welcome to your next lesson: Objects in action.

I'm setting up your lesson based on what you've learned so far, what’s coming next, and how to help you understand better.

Step by step, you’re moving forward.

Imagine you want a little digital dice machine 🎲 in Python that you can reuse many times.

In this lesson, that “machine” will be a Random object created from Python’s Random class. We’ll go slowly and keep the picture clear:

Random class → Random object → method call → produced number

First tiny idea: when you write random.Random() with parentheses, Python actually builds a new object for you — like building one dice machine.

We’ll turn that into code in a minute, but I want to check where you are:

Do you feel like you understand the difference between a class and an object/instance right now?

Yes, pretty solid
Kind of / fuzzy

Not really at all
yes

Nice 👍

Let’s make this very concrete with random.Random.

In code, we’ll do this in two steps:

Create a Random object (the “dice machine”)

```
import random
rng = random.Random()
```

Here:

Random is the class
random.Random() (with ()) creates one object
rng is a variable pointing to that object
Later, we’ll use rng to do actions (like roll a die).

Quick check before we touch methods:

Does this mapping make sense?

Random → class (blueprint)
random.Random() → creates an object
rng → refers to that object you just created

You can answer: “yes” or “I’m not fully sure”.

yes

Great 🌟

Next step: let’s make rng do something.

Think of rng as a dice machine. To “roll” it, we call one of its methods (actions) using a dot, like this:

Python

```
import random
```

```
rng = random.Random()      # 1) make the Random object
number = rng.randint(1, 6) # 2) ask it for a random int from 1 to 6
print(number)              # 3) show the produced number
Visually:



Random class → rng object → rng.randint(...) → random number



rng → the object (the machine)
rng.randint → a method (an action the machine can do)
number → the data produced by that action
Before we try this in the editor:

Is it clear that rng (the object) is not the same thing as number (the random result)?

You can say “yes” or “still a bit fuzzy”.

Yes, that’s clear

What do you think? 👇

In this code:

python
import random
rng = random.Random()
value = rng.randint(1, 10)
Which part is the object?


A.
Random


B.
random.Random


C.
rng


```

D.
rng.randint(1, 10)

E.
value

rng

Nice, you chose rng 🎯

Let’s walk through each piece step by step:

Python
import random

rng = random.Random()
number = rng.randint(1, 6)
print(number)
rng → the Random object (the dice machine) ✅
rng.randint → a method (an action the object can do)
number → the data (one random integer between 1 and 6)

So your answer “rng” fits the question: “What is rng after this code runs?” → a Random object (instance).

Now I want you to do it yourself 💻

In the code editor, write Python code that:

Imports random
Creates a Random object and stores it in a variable named rng
Uses rng.randint(1, 100) to get a random number
Stores that in a variable named result
Prints result

Then hit Run to check it, and when it looks good, hit Submit code.

I’ll check it and then we’ll do one more quick “what is object vs method vs data?” check.

Python
import random

```
my_rand = random.Random()
result = my_rand.randint(1,10)
print(f"Result: {result}")
Nice work, Ron 🎯


Step by step, you did exactly what I wanted:

Python
import random
my_rand = random.Random()        # ✅ created a Random object (the "machine")
result = my_rand.randint(1,10)   # ✅ called a method on that object
print(f"Result: {result}")       # ✅ printed the produced data
Let’s label the three key parts clearly:


my_rand → the object (a Random instance)
my_rand.randint → the method (an action that can produce random integers)
result → the data returned by that method
One last thing I want to make crystal clear for this lesson is that distinction.


Look at this mini version of your code:

Python
import random
rng = random.Random()
value = rng.randint(1, 10)
Can you tell me, in your own words (one short sentence each is enough):


What is rng?
What is rng.randint?
What is value?

You don’t need to be fancy, just clear.

rng is an object
rng.randint is a method
value is data

```

Take your time and answer the question below 👇

What is rng after this code runs?

python
import random
rng = random.Random()

A.
The Random class

B.
A Random object (instance)

C.
A number produced by a method

D.
The random module

Nice job, Ron 🎉 You nailed it.

For this code:

import random
rng = random.Random()

You chose “a Random object (instance)” — that’s exactly right ✅

Random (with a capital R) is the class (the blueprint).
random.Random() (with ()) creates a new object from that class.
rng is the variable that refers to that object — your own personal random-number machine.

You also clearly explained:

rng → object
rng.randint → method

value / result → data

That’s the full distinction this lesson wanted you to master 💪

You’ve completed Objects in action ✅

Next up, you’ll keep building on this object-oriented way of thinking.

If you want, tell me: anything still a tiny bit fuzzy about objects vs methods vs data, or does it feel solid enough to move on and just reinforce by practice?

```
import random
my_rand = random.Random()
result = my_rand.randint(1,10)
print(f"Result: {result}")
```